

Effects of Putter-Face Material and Surface Topography on Golf Ball Launch and Roll Transition

Parker Cole (A20471508), Yujiang “Mike” Xiang

School of Mechanical and Aerospace Engineering, Oklahoma State University, Stillwater, OK 74078

My research interest lies in golf club design and impact mechanics, specifically how putter-face material properties and surface topography influence a golf ball immediately after impact. I have designed and prototyped an interchangeable-face putter, and this project will use controlled testing to separate face effects from changes in putter geometry or stroke delivery. The goal is to determine which measurable face properties influence initial ball speed, spin, launch, bounce/skid behavior, transition to true roll, and repeatability. Rather than assuming that a rougher, softer, or grooved face is inherently better, the study will identify the physical mechanisms responsible for any observed differences. This work will provide practical experience in materials characterization, experimental mechanics, high-speed imaging, uncertainty analysis, and quantitative data analysis.

RESEARCH BACKGROUND

A putted golf ball generally passes through launch/bounce, skid, and true rolling phases. True roll occurs when the translational velocity of the ball equals its peripheral velocity, $v = \omega R$. During skid, turf friction changes both translational and rotational velocity until this condition is reached. Previous studies show that delivered loft, impact location relative to the putter center of gravity, mass distribution, and shaft coupling can strongly affect launch spin and skid [1,2]. Face design also matters; interchangeable face materials and grooves can change coefficient of restitution (COR), ball speed, and putt length, while controlled milling depth and pitch can produce measurable differences in ball speed [2,3]. Face compliance has also been shown to modify effective COR and distance consistency [5]. These results establish that putter-face design can alter impact behavior, but they do not fully identify which material and surface properties govern the complete transition from impact to true roll.

SCIENTIFIC GAPS AND CHALLENGES

The literature reviewed addresses the major pieces of the problem separately. Brouillette [2] showed that the tested face materials and grooves primarily changed normal impact behavior/COR; once putt length was normalized, skid distance was essentially unchanged. Lambeth et al. [3] showed that milling geometry can alter energy transfer, but did not measure initial spin, bounce, or distance to true roll. Pope et al. [4] developed a rigorous high-speed-video method for measuring skid and true roll, but varied loft rather than face material or surface condition. Emerson et al. [5] demonstrated that face compliance changes COR, while Burritt and Henrikson [6] showed that putting-surface rolling resistance itself varies with velocity and surface type. Thus, a gap remains in systematically characterizing material properties and quantitative surface topography (e.g., hardness, elastic modulus, effective COR, Ra/Rz, milling pitch/depth) and directly connecting them to the complete ball response. Major challenges are controlling delivered loft, impact height, face thickness/stiffness, putter speed, CG/MOI, ball condition, and the putting surface so that small face effects are not masked by other variables.

PROPOSED RESEARCH APPROACH

Testing will use a repeatable mechanical pendulum and one putter chassis with interchangeable face inserts. Phase I will compare selected face materials while matching face geometry, thickness, loft, and surface condition as closely as practical. Phase II will hold material constant and vary measured surface roughness and/or milling geometry. Material and surface properties will be characterized by using available metrology and testing resources, with candidate quantities including elastic modulus, hardness, effective COR, Ra, Rz, groove/milling pitch, and depth. A high-speed camera will record the ball with three high-contrast black tracking dots. The footage will be analyzed using BioDynamics Optimization Lab Software, which Dr. Xiang has experience using, to track translational and rotational motion and determine initial speed v_0 , initial angular velocity ω_0 , launch angle, bounce behavior, skid distance, and the position at which $v = \omega R$. Multiple repetitions at controlled impact speeds will establish measurement uncertainty and final-distance variability. ANOVA and/or regression analysis will be used to distinguish statistically significant material and surface effects and to determine whether changes in spin and roll transition are independent of changes in ball speed. A central outcome will be a property-to-performance relationship that can guide putter-face material selection and surface design for more predictable ball behavior.

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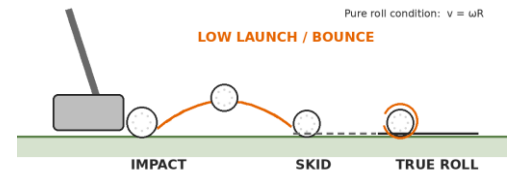


Figure 1. Conceptual ball motion following putter impact.

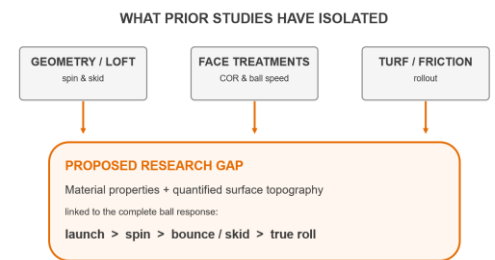


Figure 2. Existing literature areas and the proposed research gap.

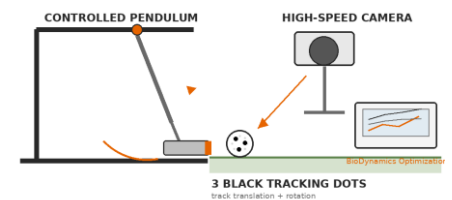


Figure 3. Proposed pendulum, high-speed camera, and three-dot motion-tracking concept.